Linear Algebra II, Sheet 2, HT26

Determinants. Eigenvalues and eigenvectors.

Main course

1. Evaluate the $n \times n$ determinant

$$\begin{vmatrix} 2 & -1 & 0 & 0 & \cdots & 0 \\ -1 & 2 & -1 & 0 & \cdots & 0 \\ 0 & -1 & 2 & -1 & \ddots & \vdots \\ 0 & 0 & -1 & \ddots & \ddots & 0 \\ \vdots & \vdots & \ddots & \ddots & 2 & -1 \\ 0 & 0 & \cdots & 0 & -1 & 2 \end{vmatrix}.$$

2. Let J be the $n \times n$ matrix all of whose entries are 1.
(i) Determine χ_J and show that there are two eigenvalues, 0 and n.
(ii) Find the corresponding eigenspaces. Is J diagonalizable?

3. Let $S\colon V \to V$ be a linear map on a vector space V such that $S^2 = I$.
(i) Show that the only possible eigenvalues of S are 1 and -1. Let U denote the 1-eigenspace and W denote the -1-eigenspace.
(ii) Suppose that $v \in V$ and $v = u + w$ where $u \in U$ and $w \in W$. Determine u and w in terms of v and Sv.
(iii) Verify that $u \in U$ and $w \in W$ for the u, w found in (ii) and deduce that $V = U \oplus W$.
(iv) If $\mathcal{B}_U$ and $\mathcal{B}_W$ are bases for U, W what is the matrix for S wrt $\mathcal{B}_U \cup \mathcal{B}_W$?
(v) Let $V = M_{n\times n}(\mathbb{R})$ and $S(A) = A^T$. Describe U and W and find an eigenbasis for S.

4. Find the characteristic polynomials of the following matrices. Are any of the matrices similar? [Hint: for no case do you need to construct a change of basis matrix that transforms one matrix into another.]

$$\begin{pmatrix} 2 & 6 & 0 \\ 1 & 1 & 0 \\ 1 & -2 & 3 \end{pmatrix}, \quad \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{pmatrix}, \quad \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -1 & 1 & 1 \end{pmatrix}, \quad \begin{pmatrix} -1 & 1 & 1 \\ 0 & 3 & 0 \\ 0 & 0 & 4 \end{pmatrix}.$$

5. Let m, n be real numbers and let a, b, c be the roots of the cubic $z^3 + mz + n = 0$. The *discriminant* Δ of the cubic is given by

$$\Delta = (a-b)^2(a-c)^2(b-c)^2.$$

(i) Explain why Δ is a real number. What can be said about the roots of the cubic when $\Delta > 0$, $\Delta = 0$, $\Delta < 0$?
(ii) By considering the identity $(z-a)(z-b)(z-c) = z^3 + mz + n$, show that $a+b+c = 0$ and find expressions for m and n in terms of a, b, c.
(iii) Let $S_k = a^k + b^k + c^k$ for $k \geqslant 0$. Find expressions for S_0, S_1, S_2 in terms of m, n.
(iv) Explain why $S_{k+3} + mS_{k+1} + nS_k = 0$ for $k \geqslant 0$ and hence find expressions for S_3 and S_4 in terms of m and n.
(v) Write down $\det A$ where

$$A = \begin{pmatrix} 1 & a & a^2 \\ 1 & b & b^2 \\ 1 & c & c^2 \end{pmatrix}.$$

Determine the matrix $A^T A$ and deduce that $\Delta = -4m^3 - 27n^2$.

Starter

S1. (i) Show that a square matrix is singular if and only if 0 is an eigenvalue.
(ii) Show that a square matrix has the same eigenvalues as its transpose.
(iii) If A is an invertible matrix, how are the eigenvalues of A^{-1} related to those of A?

S2. (i) Show that the product of two $n \times n$ permutation matrices is a permutation matrix.

(ii) Show that the inverse of a permutation matrix is a permutation matrix.

(iii) Show that a permutation matrix is orthogonal.

S3. (*Parity is well-defined*) Any permutation matrix can be written as a product of transposition matrices S_{ij}. For a given permutation matrix, show that this product always involves an even number of transpositions, or always an odd number. List the even 3×3 permutation matrices.

Pudding

P1. Let $a_1, \ldots, a_n$ be real numbers. By repeated differentiation, or otherwise, show that each a_i must be zero if

$$a_1 \sin x + a_2 \sin 2x + \cdots + a_n \sin nx = 0 \qquad \text{for all real } x.$$

P2. Let A be an $m \times n$ matrix. A *submatrix* of A is any matrix formed by deleting certain rows and columns of A. The *determinantal rank* of A is the greatest value of r such that there is an invertible $r \times r$ submatrix.

(i) For each matrix below, find its determinantal rank.

$$\begin{pmatrix} 1 & 1 & 1 \\ 2 & 2 & 2 \\ 3 & 3 & 3 \end{pmatrix}, \qquad \begin{pmatrix} 1 & 0 & 1 \\ 2 & 1 & 2 \end{pmatrix}, \qquad \begin{pmatrix} 1 & 1 & -1 & 1 \\ 2 & 3 & 0 & 2 \\ 1 & 2 & 1 & 1 \end{pmatrix}.$$

(ii) Show that determinantal rank of a matrix equals its rank.

P3. Let $A = (a_{ij})$ be an $n \times n$ matrix where $n \geqslant 2$. Show that the adjugate $\mathrm{adj}A = 0$ if and only if $\mathrm{rank}A \leqslant n - 2$. What is the rank of $\mathrm{adj}A$ if $\mathrm{rank}A = n - 1$?